

98.7 percent winning percentage. To calculate the probability of this achievement if markets were efficient, we make the simplifying assumption that the average win and average loss were equal. (This is a very conservative assumption since, in fact, Thorp's average win was significantly higher, which implies that the probability of Thorp achieving his win percentage by chance will be even lower than the estimate we derive.) Given the assumption that the average win and loss are about equal, the probability of any single trader achieving 227 winning months (or better) out of 230 is equivalent to the probability of getting 227 or more heads in 230 coin tosses, which is approximately equal to an infinitesimally small 1 out of 10^{63} . Even if we assume 1 billion traders, which is a deliberate exaggeration, the odds of getting at least one track record equivalent to or better than Thorp's would still be less than 1 out of 10^{62} . To put this probability in context, the odds of randomly selecting a specific atom in the earth would be about a trillion times better.¹

There are two ways of looking at these results:

1. Boy, Thorp was unbelievably lucky!
2. The efficient market hypothesis is wrong.

Track records such as Thorp's prove conclusively that it is possible to beat the market and that the large group of economists who insist otherwise are choosing to believe theory over evidence.² The contention that it is possible to beat the markets, however, does not say anything about the difficulty of the task. In fact, it is the difficulty in beating the market (the vast majority of market participants fail to do so) that helps create the illusion that markets are efficient.

Thorp's career encompasses an extraordinary number of first achievements:

- He co-developed (along with Claude Shannon) the first wearable computer that could be used to win at roulette.
- He developed the first blackjack betting strategy that provided a positive edge to the player, which he divulged in his global best seller, *Beat the Dealer*. The book changed the way casinos operate.
- Thorp along with Sheen Kassouf developed the first known systematic approach to trading warrants and other convertible securities (e.g., options, convertible bonds, convertible preferred stocks) by hedging